

Card (1995) — Instrument & Weight Diagnostics

Why the estimates look the way they do: Blocks B, E, and G explained with numbers

This note works through the three diagnostic blocks from `card_descriptive_analysis.R` and explains, with the actual numbers, why the weight diagnostics in Section 5.1 — $ESS \leq 4$, ~33% negative weights, $Max|\omega|$ up to 0.277 — are not surprising once you look at the instrument and sample structure. Each finding connects directly to a specific feature of the data.

Block B — Instrument Strength: Two Very Different First Stages

The instrument `nearc4` (grew up near a four-year college) predicts both treatment definitions, but with radically different strength. Table B.1 shows the raw 2×2 cross-tabulations and first-stage differences for D1 (some college, `educ > 12`) and D2 (college completion, `educ ≥ 16`).

Table B.1: Raw first-stage cross-tabulations by treatment definition

	D1 = 0 (no some-college)	D1 = 1 (some college)	Total
Z = 0 (not near college)	553	404	957
Z = 1 (near college)	936	1,117	2,053
Total	1,489	1,521	3,010

	D2 = 0 (no completion)	D2 = 1 (college completion)	Total
Z = 0 (not near college)	742	215	957
Z = 1 (near college)	1,451	602	2,053
Total	2,193	817	3,010

The first-stage difference — $P(D=1|Z=1) - P(D=1|Z=0)$ — is the denominator of every IV estimator. Table B.2 summarises both definitions with and without controls.

	D1 (somecol)	D2 (educ16)	Ratio D1/D2
$P(D=1 Z=1)$	0.5441	0.2932	1.86
$P(D=1 Z=0)$	0.4222	0.2247	1.88
First-stage diff (no controls)	0.1219	0.0686	1.78
F-statistic (no controls)	39.30	15.59	2.52×
First-stage coeff (Card controls)	0.0636	0.0302	2.10×
t-statistic (Card controls)	3.44 ***	1.79 .	

Table B.2: First-stage strength by treatment definition. Green = acceptable; amber = borderline weak; red = problematic. Significance: . $p < 0.10$, *** $p < 0.001$.

What this means in practice.

Every IV estimator can be written as $\tau = ITT(Y) / ITT(D)$, where $ITT(D)$ is the first-stage difference. In the outcome-weights framework, this division propagates directly into the weight vector: $\omega_i \propto 1/ITT(D)$. So a first stage of 0.0302 (D2 with controls) means every weight is scaled up by a factor of roughly

$1/0.0302 \approx 33$ relative to what it would be with a first stage of 1. The ESS, which is inversely proportional to $\sum \omega_i^2$, therefore collapses by a factor of $33^2 \approx 1,000$. This is the arithmetic explanation for ESS = 1 in the educ16 cells — not a modelling failure, but a consequence of the instrument generating almost no variation in college completion conditional on the covariates.

Crucially, **the overlap diagnostics for $P(D2=1|X)$ show no extreme propensity scores**: the range is [0.07, 0.44] with no observations below 0.05 or above 0.95. So the ESS collapse for the educ16 cells is *not* a standard overlap problem that propensity trimming could solve. It is a weak-instrument problem: the denominator of the Wald ratio is small, amplifying all weights uniformly. The solution is a stronger instrument, not better reweighting.

Block E — Propensity Score Distributions: Why ~33% Negative Weights

The negative weight share (~31-33% across all estimators and cells) is not an artefact of any particular estimator — it is a mechanical consequence of the instrument propensity score distribution. Table E.1 shows $P(Z=1|X)$ for both covariate specifications.

Statistic	Card spec (14 covariates)	Kitagawa spec (5 covariates)
Min	0.190	0.253
Q1	0.430	0.455
Median	0.791	0.798
Mean	0.682	0.682
Q3	0.874	0.878
Max	0.950	0.907
Std dev	0.2355	0.2280
% below 0.30	7.9%	7.3%
% above 0.70	63.9%	63.8%
Implied ESS ($\tau_{ml,u}^2, D1$)	4.63	2.94

Table E.1: $P(Z=1|X)$ instrument propensity score distribution. Both specs produce near-identical distributions despite very different numbers of covariates.

The right-skew problem: 64% of the sample has $p_{\blacksquare} > 0.70$.

The median instrument propensity is 0.791 — meaning the typical observation lives in an area that is 'almost always near a college'. This right-skew is the structural cause of the negative weight share. To see why, recall the kappa weight formula for the unnormalised estimator:

$$\omega_i \propto Z_i/p_{\blacksquare_i} - (1-Z_i)/(1-p_{\blacksquare_i})$$

For a $Z=0$ observation with the median propensity $p_{\blacksquare} = 0.791$, this gives:

$$\omega_i \propto 0/0.791 - 1/(1-0.791) = 0 - 1/0.209 = -4.78$$

That is a large negative weight — not because this observation is unusual, but because it is the *typical* $Z=0$ observation in this dataset. Table E.2 shows the kappa weight sign and magnitude for representative propensity score values, illustrating how the weight structure follows mechanically from the instrument distribution.

$p(Z=1 X)$	Z=1 obs. ω_∞	Z=0 obs. ω_∞	Sign (Z=0)	Interpretation
0.20 (low)	+5.00	-0.25	neg. (small)	7.9% of sample
0.43 (Q1)	+2.33	-1.75	negative	below median Z=0
0.50 (equal)	+2.00	-2.00	negative	perfectly balanced Z
0.79 (median)	+1.27	-4.78	large neg.	typical obs. in data
0.87 (Q3)	+1.15	-7.69	very large neg.	63.9% above 0.70
0.95 (near max)	+1.05	-19.0	extreme neg.	tail observations

Table E.2: Kappa weight direction and magnitude for representative propensity score values. Row highlighted in amber = the typical (median) observation in the Card (1995) dataset. All Z=0 observations receive negative weights; the magnitude grows sharply as $p(Z=1|X) \rightarrow 1$.

Why Card and Kitagawa specs give different ESS despite identical propensity distributions.

The two specifications produce almost the same $P(Z=1|X)$ distribution: same mean (0.682), similar SD (0.235 vs 0.228), similar tail shares (Table E.1). Yet the implied ESS differs substantially: **4.63 under Card vs. 2.94 under Kitagawa**. This seems paradoxical — if the instrument propensity is the same, why does the ESS differ?

The answer is that the kappa weight ω_i depends on *both* $p(Z=1|X_i)$ and the treatment D_i . The Kitagawa specification uses fewer covariates to partial out the treatment, which means the residual variation in D after removing X effects is less well explained. This increases the variance of the effective Wald denominator across observations, concentrating the identification weight on fewer units and reducing the ESS — even though the instrument propensity looks identical. In short: **the Kitagawa spec is worse not because it predicts Z badly, but because it predicts D less precisely**, which makes the Wald denominator noisier across observations.

The $P(D2|X)$ result and what it actually means. The treatment propensity for D2 (college completion) under Card controls ranges from 0.067 to 0.440, with no observations below 0.05 or above 0.95. This means there is *no overlap problem* for the educ16 treatment: the AIPW augmentation term has no near-zero denominators, and propensity trimming would not help. The $\text{Max}|\omega| = 0.277$ for Wald-AIPW/educ16 comes entirely from the weak first stage (0.030 conditional on controls), which scales up all weights by a factor of ~33. This is a qualitatively different fragility than overlap failure — and the distinction matters because the remedies are different: better propensity estimation cannot fix a weak instrument.

Block G — First-Stage Heterogeneity: Why Region Dummies Dominate the Love Plots

Even if the overall first stage is acceptable, IV estimates are only well-identified if the instrument's effect on treatment is approximately uniform across the sample. When the first stage varies sharply across subgroups, the kappa reweighting over-represents high-compliance groups and under-represents low-compliance ones, creating imbalance on any covariate that correlates with those groups. Table G.1 shows this heterogeneity for the eight Census regions.

Region	N	fs D1	fs D2	Ratio D1/overall	Interpretation
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reg662	484	0.217	0.199	1.78×	78% above average — compliers concentrated here
reg665	627	0.122	0.040	1.00×	Representative — close to overall average
reg663	589	0.098	0.050	0.81×	Slightly under-represented
reg661	140	0.061	0.136	0.50×	Half the average D1 response
reg664	193	0.066	0.118	0.54×	Half the average D1 response
reg667	331	0.091	−0.051	0.74×	D2 negative — proximity hurts completion here
reg666	289	0.024	−0.001	0.20×	Near-zero D1; D2 ≈ 0 — almost no compliers
reg668	85	−0.161	−0.073	−1.32×	NEGATIVE — local monotonicity violation
Overall	3,010	0.122	0.069	1.00×	Reference

Table G.1: First-stage heterogeneity across Census regions. $fs\ D1 = P(D1=1|Z=1) - P(D1=1|Z=0)$ within region. Ratio = region first stage / overall first stage. Green = above average; amber = substantially below; red = problematic. Rows sorted by D1 ratio descending.

Three distinct problems revealed by the regional breakdown.

1. Extreme concentration in reg662 (ratio = 1.78). This single region, containing 484 observations (16% of the sample), drives a disproportionate share of the complier population. The kappa reweighting effectively up-weights reg662 by 1.78× relative to its sample share. Any covariate correlated with living in reg662 — geography, industry structure, local labour market conditions — will show imbalance in the Love plots. This is not a failure of the estimator; it is a consequence of the instrument being more 'active' in some regions than others.

2. Near-zero compliance in reg666 (ratio = 0.20) and reg664/reg661 (ratio ≈ 0.50). In reg666, the first stage for D1 is only 0.024 — college proximity barely affects whether men in this region attend some college. These observations contribute almost nothing to the first stage, so they are effectively excluded from identification but remain in the sample, receiving small weights that add noise. For D2, reg666 has a first stage of −0.001 — essentially zero — meaning the educ16 instrument is completely uninformative for this region. The educ16 LATE is identified almost entirely off reg662 and a few other regions.

3. Negative first stage in reg668 (ratio = −1.32). In region 668 (N=85), the first-stage difference is −0.161 for D1 and −0.073 for D2. Being near a college is associated with *less* college attendance in this region. This is a local violation of the monotonicity assumption (no defiers) that underlies all LATE identification. While 85 observations is a small share of the sample, this finding means the instrument is not monotone in the full sample — the LATE interpretation requires an approximation that monotonicity holds 'on average', which is a weaker claim than it appears.

How first-stage heterogeneity produces Love plot failures.

The mechanism is direct. Define the effective weight for observation i as $\omega_i \propto (Z_i/p_{11} - (1-Z_i)/(1-p_{11})) / \int (Z-p_{11})D\ p$. The denominator is the first stage. When the first stage is 1.78× the average in reg662, observations in that region receive 1.78× the weight they would have under uniform compliance. Conversely, reg666 observations receive only 0.20× the average weight.

Table G.2 illustrates the implied weight scaling by region for the D1 somecol estimator.

Region	N	Sample share	Effective weight scale (D1)	Implied complier share
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reg662	484	16.1%	1.78×	~28.7% of compliers
reg665	627	20.8%	1.00×	~20.8% of compliers
reg663	589	19.6%	0.81×	~15.9% of compliers
reg666	289	9.6%	0.20×	~1.9% of compliers
reg668	85	2.8%	-1.32×	negative weight

Table G.2: Implied weight scaling and complier share by region. Weight scale = region first stage / overall first stage. Complier share = sample share × weight scale (approximate). Selected regions only; full table in Block G output.

The Love plot implication. A Love plot measures whether the weighted covariate means are balanced between treated and control. When reg662 observations carry 1.78× their natural weight, the weighted mean of every covariate correlated with reg662 will be shifted toward the reg662 value. If reg662 is, say, an urban Northeast region with high baseline wages and high college density, then covariates like *smsa*, *south*, and regional indicators will show imbalance in the Love plot. This is exactly the pattern seen in Block F of `card_22_double_ml.R`: region dummies are the covariates that most frequently exceed the $|\text{SMD}| \leq 0.1$ threshold.

Synthesis: Three Facts That Explain All the Weight Diagnostics

The three blocks tell a single coherent story. The weight diagnostic pattern in Section 5.1 — $\text{ESS} \leq 4$, ~33% negative weights, $\text{Max}|\omega|$ up to 0.277 — follows entirely from these three features of the Card (1995) dataset.

Diagnostic pattern	Root cause (Block)	Mechanism	Remedy
~33% negative weight share	Right-skewed $p_{\text{D2}}(Z=1 X)$ (Block E)	64% of sample has $p_{\text{D2}} > 0.70$. A $Z=0$ obs. with $p_{\text{D2}}=0.79$ gets kappa weight -4.78. All $Z=0$ obs. receive negative weights; their share (~33%) is determined by the instrument take-up rate.	None — structurally unavoidable given this instrument and sample.
$\text{ESS} \leq 4$ (somecol)	Weak conditional first stage (Block B)	First stage drops from 0.122 (no controls) to 0.064 (with controls). Weights scale by $1/0.064 \approx 16\times$, collapsing the ESS.	None without a stronger instrument. Propensity trimming would not help.
$\text{ESS} = 1$ (educ16)	Near-zero conditional fs (Blocks B+E)	Conditional first stage for D2 is 0.030 ($t=1.79$). Weights scale by $1/0.030 \approx 33\times$. No overlap problem: $P(D2 X)$ has no extremes. Weak instrument, not weak overlap.	A stronger instrument for college completion specifically.
$\text{Max} \omega $ high for Wald-AIPW	First-stage amplification (Block B)	Wald-AIPW augments IPW with an outcome model. When the first stage is near zero, the AIPW correction amplifies the most informative observations (those in reg662) even further. Kappa weights avoid this augmentation and achieve lower $\text{Max} \omega $.	Use normalised kappa estimators, which do not augment with an outcome model correction.
Love plot failures on region dummies	First-stage heterogeneity (Block G)	reg662 has 1.78× the average first stage; reg666 has 0.20×; reg668 has a negative first stage. Compliers are geographically concentrated, making the weighted sample unrepresentative on region.	No statistical fix. A geographically broader instrument would be needed.

Table S.1: Summary of diagnostic patterns, root causes, and implications. All five patterns in the Section 5.1 weight diagnostics trace to instrument properties, not to estimator failure.

Note: All numbers in this document are computed from the Card (1995) dataset (N=3,010) using `card_descriptive_analysis.R`. Propensity scores estimated via logit MLE. ESS = $1/\sum_i w_i^2$. Kappa weights via `kappa_outcome_weights()`.